

WASTEWISE SCAN · DESIGN DECISIONS REGISTER

WasteWise Scan design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-02	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

WasteWise Scan design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P12 (shell screws and gasket, window clamp, shroud flange, optical head block, photodiode seat, cradle and strap, USB-C seat, display frame, button hole, slide switch, hood clips, calibration cap)	Accept; accept with changes; reject	Accept	The whole build plan	WSC-DDR-004, A2
2	Mass (R6) is now not met: 274 g with the cap, 254 g without, against 250 g	(a) shell walls 2.0 mm instead of 2.5 mm, saving about 20 g, with the drop case checked at TRL 4; (b) count R6 without the cap; (c) accept about 275 g and restate R6	(a), then weigh the first prototype before (b) or (c)	Shell print settings and the making sketches of both shells	WSC-DDR-004, A1; WSC-CAL-001 [A3]
3	First partner and region for co-design and the first price table	Partners are picked per area later	None yet	Not part of the TRL 3 build; sets the price table in the firmware	WSC-DDR-001, O1
4	Who the USD 150 value-engineering target is for: cooperatives or individual pickers	Cooperatives; individual pickers	None yet	Not part of the build	WSC-DDR-001, O2
5	An extended InGaAs band (1,700 to 1,750 nm) to separate PE from PP	Add a ninth band on an extended InGaAs detector; keep eight bands	None yet; a paper band study on published reference spectra comes first	The LED set, the photodiode and the head block pockets	WSC-DDR-001, O3
6	How the pair ID that joins a scan to a WasteWise-ml photo is created in the field	Typed digits; a sticker; the phone scanning a code on the scanner screen	None yet	Firmware only	WSC-DDR-002, question 2
7	Whether consented spectra join the WasteWise-ml dataset, and on what license and ownership terms	To be set with the WasteWise-ml side	None yet	Not part of the build	WSC-DDR-002, question 3

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The protected cell's length (the cradle allows 69 mm) and the contact pair's thickness and spring travel	The cradle end walls are 75 mm apart for a 69 mm cell, a 1 mm flat contact and a spring compressed to 4 mm plus its 1 mm plate	WSC-DDR-004, P6
2	The display board's real outline (about 62 x 26 x 9.5 mm is assumed), its connector positions and its battery input	The display frame and the 0.25 mm gap to the ceiling are set from the assumed outline; the slide switch is wired in the battery lead	WSC-DDR-004, P8
3	Whether the board charges the cell with the switch in the battery lead, and from its 5 V pin	Charging needs the switch on; if the board cannot charge through its 5 V pin, the USB-C board wiring changes	Build plan, wiring
4	The button's bezel (19 mm or less), its depth under the panel (14 mm or less) and its nut (20 mm or less across)	They set the button hole and its clearance to the display frame and the amplifier board	WSC-DDR-004, P9
5	The slide switch's mounting hole pitch (15 mm) and body size (10 x 6 x 6 mm)	The wall slot and holes are printed into the bottom shell	WSC-DDR-004, P10
6	Each LED's package: TO-46 (5.4 mm) or 5 mm lens LED (5.8 mm rim)	The head block pockets are 5.6 mm, for TO-46 cans	WSC-DDR-004, P4
7	The foam gasket squeezes to about 0.6 mm with the bosses touching	The bosses stop the squeeze at 0.6 mm; a harder foam will not seal	WSC-DDR-004, P1
8	The photodiode's responsivity curve near 1,700 nm	It decides how much of the 1,650 nm band is seen (R1)	BOM line 4; WSC-CAL-001 B

■ Value engineering

Value-engineering target: USD 150 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 167.50 (USD 17.50 over the target). Main cost drivers and savings worth trying:

- The largest lines are the eight LEDs with their drivers (USD 62; the six short-wave infrared LEDs cost about USD 8 to 15 each), the InGaAs photodiode (USD 30) and the display board (USD 20). Optics are 55 % of the cost.
- Making the design constructable added USD 3.50 net: the display frame, cell strap and gasket (lines 17 to 19) and more fixings; the bought cell holder became contacts in a printed cradle, which saved USD 1.
- Savings worth trying: the six-band option (dropping 1,050 and 1,300 nm saves about USD 18) if the band study shows six bands suffice (WSC-DDR-001 D2); buying the LEDs as a set from one maker; a second source for the photodiode; at volume, a bare ESP32-S3 module and display in place of the display board.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D7: discrete LEDs with an InGaAs photodiode; USD 150 kept as the volume target with about USD 163 accepted for the prototype; T-Display-S3 class board; the eight-band set; a small on-device classifier with shared data; PTFE white reference in the cap; one protected 18650 cell	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	WSC-DDR-001
2026-09-25	D8 to D12: back clear items in sun with the black cap and refuse a result on through-light; black calibration cap; clip-on sun hood; open-air crosstalk reading in calibration; one shared scan record with a pair ID join	Amish: "i accept all your recommendations, go with them across all repos."	WSC-DDR-003, WSC-DDR-002
2026-09-26	WasteWise Scan chosen as the pilot for product-grade renders; every drawing names the project and its repository	Amish	docs/ REVIEW.md , session 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan; outstanding decisions go in this register, not the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	WSC-DDR-004 (changes open for review, item 1 above)
2026-10-01	budget_usd is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register, Value engineering